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 EXPONENT LAWS

Multiplying	$a^m \cdot a^n = a^{m+n}$
Dividing	$\frac{a^m}{a^n} = a^{m-n}$
Power of a power	$(a^m)^n = a^{mn}$
Negative exponent	$a^{-n} = \frac{1}{a^n}$
Zero exponent	$a^0 = 1, \quad a \neq 0$
Root as a power	$\sqrt[n]{a} = a^{\frac{1}{n}}$
These only combine powers of the same base. $2^3 \cdot 3^2$ does not simplify — rewriting both sides to a common base is usually the first step the question is testing.	

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 EQUATIONS & INEQUALITIES

Quadratic (factorised)	$(x - p)(x - q) = 0 \Rightarrow x = p \text{ or } x = q$
Simultaneous	substitute one equation into the other, or add/subtract to eliminate
Inequality — divide by a negative	reverse the sign
Interval notation	$x \in [a; b)$
A quadratic must equal zero before you factorise it. Moving everything to one side is not tidying up — it is the step that makes the method valid.	

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 FUNCTIONS & GRAPHS

Function	Shape	Key features
$y = ax + q$	straight line	gradient a · y -intercept q
$y = ax^2 + q$	parabola	turning point $(0; q)$ · $a > 0$ opens up
$y = \frac{a}{x} + q$	hyperbola	asymptotes $x = 0$ and $y = q$
$y = ab^x + q$	exponential	asymptote $y = q$ · y -intercept $a + q$

Domain is the x -values the graph uses; range is the y -values it reaches. For the hyperbola and the exponential the asymptote is exactly the value the range excludes.

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 TRIGONOMETRY

In a right triangle

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}}$$

$$\tan \theta = \frac{\text{opp}}{\text{adj}}$$

θ	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
tan	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined

SOH-CAH-TOA only works in a right-angled triangle. Without a right angle you need the sine or cosine rule, which arrive in Grade 11.

x
 FACTORISING

Difference of squares	$a^2 - b^2 = (a - b)(a + b)$
Trinomial	$x^2 + (p + q)x + pq = (x + p)(x + q)$
Sum of cubes	$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
Difference of cubes	$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
Grouping	$ax + ay + bx + by = (a + b)(x + y)$
Take out the common factor first, every time. Half the "hard" factorising questions become one of the standard forms the moment the common factor is gone.	

Σ
 LINEAR NUMBER PATTERNS

General term	$T_n = a + (n - 1)d$
Common difference	$d = T_2 - T_1 = T_3 - T_2$
Check the difference between at least two pairs before trusting it. If it is not constant the pattern is not linear, and $T_n = a + (n - 1)d$ does not apply.	

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 ANALYTICAL GEOMETRY

Distance	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Midpoint	$M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$
Gradient	$m = \frac{y_2 - y_1}{x_2 - x_1}$
Distance uses squares, so the order of the points never matters. Gradient does not — swap the points in the top and the bottom together, or the sign comes out wrong.	

R FINANCE

Simple interest	$A = P(1 + in)$
Compound interest	$A = P(1 + i)^n$
P	the amount at the start
i	rate as a decimal (8% $\rightarrow$ 0,08)
n	number of periods
Interest compounded monthly at 12% a year means $i = 0,01$ and n counts months. Divide the rate and multiply the periods by the same number.	

M MEASUREMENT

Rectangular prism	$V = l \times b \times h$
Any right prism	$V = \text{base area} \times h$
Cylinder — volume	$\pi r^2 h$
Cylinder — surface area	$2\pi r^2 + 2\pi r h$
Circle	$A = \pi r^2, \quad C = 2\pi r$
Multiplying every dimension by k multiplies the area by k^2 and the volume by k^3 — worth learning now, because Grade 11 examines it directly.	

Σ STATISTICS

Mean	$\bar{x} = \frac{\sum x}{n}$
Range	$\text{max} - \text{min}$
Interquartile range	$Q_3 - Q_1$
Five-number summary	$\text{min} \cdot Q_1 \cdot \text{median} \cdot Q_3 \cdot \text{max}$
Order the data before finding a median or a quartile. Every mark lost on this topic is lost to an unsorted list.	

P PROBABILITY

Probability	$P(A) = \frac{\text{favourable}}{\text{total}}$
Complement	$P(\text{not } A) = 1 - P(A)$
Addition rule	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
Mutually exclusive	$P(A \text{ and } B) = 0$
Relative frequency	$\frac{\text{times it happened}}{\text{times you tried}}$
Subtracting $P(A \text{ and } B)$ is not optional — without it the overlap is counted twice, which is the single most common error in this section.	

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